Compiled Solutions – Intermediate Division

1. 01-02 C1 Computer Number Systems

BED converts to $1011\ 1110\ 1101_2$. Regrouping to sets of 3 digits gives $101\ 111\ 101\ 101_2$. This converts to 5755_8

5755_8 or 5755

2. 01-02 C1 Computer Number Systems

The base 8 addition gives 103327_8. Converting to base 2 gives $001\ 000\ 011\ 011\ 010\ 111_2$. Regrouping using sets of 4 digits gives $1000\ 0110\ 1101\ 0111_2$. This converts to $86D7_{16}$.

$86D7_{16}$ or 86D7

3. 01-02 C2 Computer Number Systems

$X_2 = A12_{16} - 567_8$ can be solved in either base 8 or 16.
$A12_{16} = 101000010010_2 = 101\ 000\ 010\ 010_2 = 5022_8$
So $5022_8 - 567_8 = 4233_8 = 100\ 010\ 011\ 011_2$

6

4. 02-03 C1 Computer Number Systems

$24_8 = 010100_2$
$372_{16} = 001101110010_2$

001110000110_2

5

5. 02-03 C1 Computer Number Systems

$2_8 X_8 = 4_{16} * AB_{16}$
$2_8 X_8 = 4_{10} * 171_{10}$
$2_8 X_8 = 684_{10}$
$2_{10} X_8 = 684_{10}$
$X_8 = 342_{10}$
$X_8 = 526_8$

526_8

6. 02-03 C2 Computer Number Systems

$(764_8)^2 = (500_{10})^2 = 250000_{10} = 750220_8$

750220_8

7. 03-04 C1 Computer Number Systems

A. $42_8 - 12_8 = 11000$
B. $3A_{16} + 10B_{16} = 101000101$
C. $10001_2 + 11101_2 = 101110$
D. $A_{16} + 10_8 + 10_2 = 10100$

C

8. 03-04 C1 Computer Number Systems

The equation translates as follows: 5
$8X + 10 = 27 + 23$
$8X + 10 = 50$
$8X = 40$
$X = 5$

9. 03-04 C2 Computer Number Systems

$.3A_{16} = \frac{3}{16} + \frac{10}{256} = \frac{48+10}{256} = \frac{29}{128}$ $\frac{29}{128}$

10. 04-05 C1 Computer Number Systems

$327_8 + X_{16} = 1101110110_2$ $29F_{16}$
$1\ 101\ 110\ 110_2 = 1566_8$
$X_{16} = 1566_8 - 327_8$
$X_{16} = 1237_8 = 10\ 1001\ 1111_2 = 29F_{16}$

11. 04-05 C1 Computer Number Systems

$1_8 = 1$, $100_2 = 4$, $11_8 = 9$, $10_{16} = 16$ so the sequence is 144_8
1, 4, 9, 16 … n^2 . The 10th term is $10^2 = 100 = 144_8$

12. 04-05 C2 Computer Number Systems

$FF_{16} * 10001_2 = (100_{16} - 1_{16}) * 11_{16} = 1100_{16} - 11_{16} = 10EF_{16}$ 10357_8
$1\ 000\ 011\ 101\ 111_2 = 10357_8$

13. 05-06 C1 Computer Number Systems

A. $53_{16} = 83$ C or 124_8
B. $1010001_2 = 81$
C. $124_8 = 84$
D. $82_{10} = 82$

14. 05-06 C1 Computer Number Systems

$213_8 + 10110_2 - 2A_{16} = 10001011 + 10110 - 101010 =$ 77_{16}
$111\ 0111 = 77_{16}$

15. 05-06 C2 Computer Number Systems

A. $231_{16} = 10\ 0011\ 0001$ B. $73_8 = 111\ 011$ C, B, A, D
C. $10111011_2 = 10111011$ D. $84_{10} = 1010100$

16. 06-07 C1 Computer Number Systems

Convert $A2B8C4D6_{16}$ to binary first. 24256142326_8
$A2B8C4D6_{16} = 1010\ 0010\ 1011\ 1000\ 1100\ 0100\ 1101\ 0110_2$
Then regroup and convert to octal
$10\ 100\ 010\ 101\ 110\ 001\ 100\ 010\ 011\ 010\ 110 = 24256142326_8$

17. 06-07 C1 Computer Number Systems

$10_{10} = 12_8$ and $100_{10} = 144_8$ 11
The octal palindromes between them are 22, 33, 44, 55, 66, 77, 101, 111, 121, 131, and 141.

18. 06-07 C2 Computer Number Systems

$A37E_{16} = 1010001101111110 \Rightarrow$ 10 1's 1
$67541_8 = 110111101100001 \Rightarrow$ 9 1's

19. 07-08 C1 Computer Number Systems

$155555_8 = 1\ 101\ 101\ 101\ 101\ 101_2$ 11 1's 133337_8 or B
$133337_8 = 1\ 011\ 011\ 011\ 011\ 111_2$ 12 1's
$145547_8 = 1\ 100\ 101\ 101\ 100\ 111_2$ 10 1's
$156464_8 = 1\ 101\ 110\ 100\ 110\ 100_2$ 9 1's
$171442_8 = 1\ 111\ 001\ 100\ 100\ 010_2$ 8 1's

20. 07-08 C1 Computer Number Systems

$243_8 = 163$ $91_{16} = 145$ $82_{16} = 130$ $1101_2 = 13$ 100101010_2 or
So the expression in base 10 is $163 + 145 - 130 / 13 = 298$ 100101010
and $298 = 100101010_2$

21. 07-08 C2 Computer Number Systems

$1F_{16} * X_{16} + 476_8 = 1010110010_2$ C or C_{16}
Converting to base 10, the equation becomes:
$31 * X + 318 = 690$
$31 * X = 372$
$X = 12$ Convert 12 to hex which is C.

22. 08-09 C1 Computer Number Systems

$FE94A_{16} = 1111\ 1110\ 1001\ 0100\ 1010_2 \Rightarrow$ 12 1's
$7534201_8 = 111\ 101\ 011\ 100\ 010\ 000\ 001_2 \Rightarrow$ 10 1's

2

23. 08-09 C1 Computer Number Systems

$13 = 1101$ $F4_{16} = 11110100$ $23_8 = 10011$
$1101 + 11110100 - 10011 + 10101 = 100000011$
$1\ 0000\ 0011_2 = 103_{16}$

103_{16} or 103

24. 08-09 C2 Computer Number Systems

$4_{16} * X - E8_{16} = C8_{16}$
$4 * X - 232 = 200$
$4 * X = 432$
$X = 108 = 6C_{16}$

$6C_{16}$ or 6C

25. 09-10 C1 Computer Number Systems

$23_8 = 19$
$1101_2 = 13$
$A1_{16} = 161$
$111110_2 = 62$
$1F_{16} = 31$

So $23_8 * 1101_2 + A1_{16} - 111110_2 / 1F_{16}$
$= 19 * 13 + 161 - 62 / 31$
$= 247 + 161 - 2$
$= 406_{10}$
$= 196_{16}$

196 OR 196_{16}

26. 09-10 C1 Computer Number Systems

$2A_{16} = 42$
$101110_2 = 46$
$27_8 = 23$
$250_{10} = 250$

So $2A_{16} + X_8 = 101110_2 / 27_8 + 250_{10}$
becomes $42 + X_8 = 46/23 + 250$
$X_8 = 2 + 250 - 42 = 210_{10}$
$\therefore \quad X_8 = 322_8$

322 or 322_8

27. 09-10 C2 Computer Number Systems

$10011 + 11001 = 101100$
$101100 + 001101 = 111001$
$111001 + 100111 = 1100000$
$1100000 + 0000011 = 1100011$

1100011

28. 10-11 C1 Computer Number Systems

$176_8 + 10110_2 / B_{16} + 54_{16} * 1011_2$
$= 126 + 22 / 11 + 84 * 11$
$= 126 + 2 + 84 * 11 = 1052 = 41C_{16}$

41c OR $41C_{16}$

29. 10-11 C1 Computer Number Systems

$111_8 = 73$ $3B_{16} = 59$ $111110_2 = 62$ 4
$112_8 = 74$ $58_{16} = 88$ So $88 - 59 = 29 = 11101_2$

30. 11-12 C1 Computer Number Systems

$576_{16} = 0101\ 0111\ 0110$ 7 1's CCF or CCF_{16}
$B2C_{16} = 1011\ 0010\ 1100$ 6 1's
$28F_{16} = 0010\ 1000\ 1111$ 6 1's
$CCF_{16} = 1100\ 1100\ 1111$ 8 1's

31. 11-12 C1 Computer Number Systems

$13_8 = 1*8 + 3 = 11$ 241 or 241_8
$A3_{16} = 10*16 + 3 = 163$
$1101_2 = 8 + 4 + 1 = 13$
$13_8 + A3_{16} - 1101_2 = 11 + 163 - 13 = 161 = 241_8$

32. 12-13 C1 Computer Number Systems

$2A3_{16} = 512 + 160 + 3 = 675$ 678 or 678_{10}
$1243_8 = 512 + 128 + 32 + 3 = 675$
$678_{10} = 678$
$1010100101_2 = 512 + 128 + 32 + 4 + 1 = 677$

33. 12-13 C1 Computer Number Systems

$101101_2 = 32 + 8 + 4 + 1 = 45$ 20053 or 20053_8
$B7_{16} = 176 + 7 = 183$
$101101_2 * B7_{16} = 45 * 183 = 8235 = 20053_8$

34. 13-14 C1 Computer Number Systems

$110111_2 = 55$ Sum = 55 + 63 + 58 + 84 = 260 101 or 101_8
$63_{10} = 63$ Average = $65 = 101_8$
$3A_{16} = 58$
$124_8 = 84$

35. 13-14 C1 Computer Number Systems

178_{16} = 101111000 5 1's 567 or 567_8
567_8 = 101110111 7 1's
101110110_2 = 101110110 6 1's
565_8 = 101110101 6 1's
377_{10} = 101111001 6 1's

36. 14-15 C1 Computer Number Systems

$15_8 + B1C_{16} - 100110_2 = 13 + 2844 - 38 = 2819 = B03_{16}$ B17 or $B17_{16}$
$AB8_{16} + 127_8 + 1000_2 = 2744 + 87 + 8 = 2839 = B17_{16}$

37. 14-15 C1 Computer Number Systems

101001001111 A54 or $A54_{16}$

$A47_{16} = 101001000111$		101001010000
Start counting:	101001001011	101001010001
101001001000	101001001100	101001010010
101001001001	101001001101	101001010011
101001001010	101001001110	10100**1010**00 = $A54_{16}$

38. 15-16 C1 Computer Number Systems

A. $25_8 + 1C_{16} = 10101_2 + 11100_2 = 110001_2$	3 1's	C
B. $A0_{16} - 100110_2 = 10100000_2 - 100110_2 = 1111010$	5 1's	
C. $B8_{16} + 127_{10} = 10111000_2 + 1111111_2 = 100110111_2$	6 1's	
D. $111_8 + 1000_2 = 1001001_2 + 1000_2 = 1010001_2$	3 1's	

39. 15-16 C1 Computer Number Systems

$101010_2 = 42$
$132_8 = 90$
$33_{16} = 51$
$1010100_2 = 84$
$44_{16} = 68$

$42 + 90 + 51 + 84 + 68 = 335$
$335 / 5 = 67 = 103_8$

103 or 103_8